

GALLANT IAS ACADEMY
PRELIMS CUM MAINS BATCH 2027
ECONOMY
MODULE 5
INFLATION

It refers to a **steady rise in the prices of various goods and services** in the economy over a period of time. It is measured through the **Price Index**. Inflation causes a **decrease in purchasing power**.

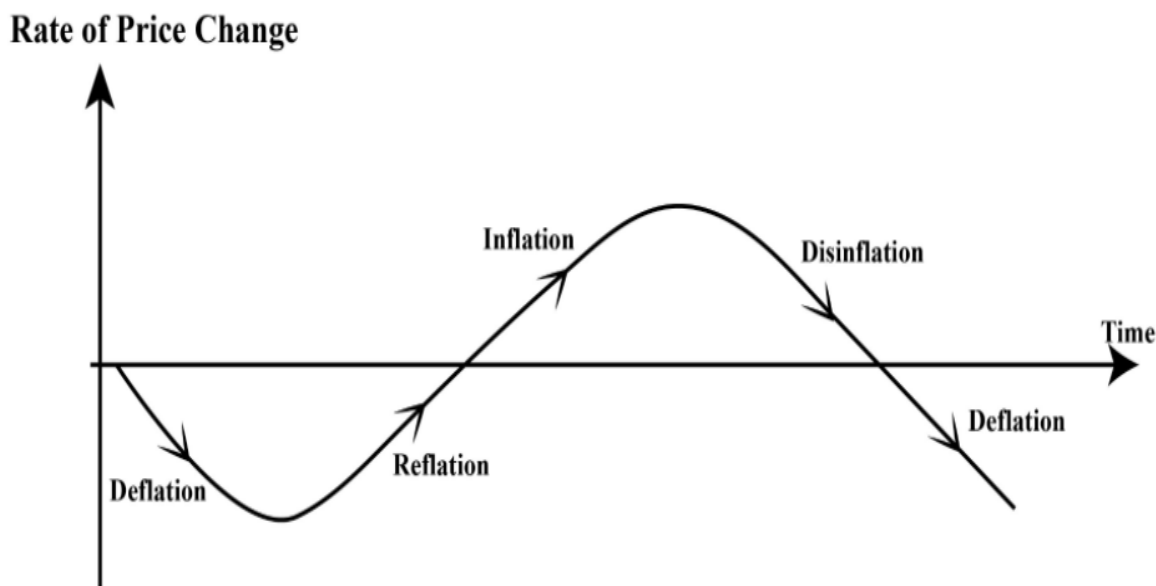
$$\text{Rate of Inflation} = (\text{Current Price} - \text{Base Year Price}) / \text{Base Year Price} \times 100$$

In the words of famous economist **Milton Friedman**, "Inflation is too much money chasing too few goods" which implies that the whole demand for the commodities is greater than the total supply.

Headline Inflation: Measures the total price change across an entire basket of goods and services, including volatile components like food and energy.

Core Inflation: Excludes these volatile items to reveal the underlying, more stable inflation trend in an economy.

RELATED CONCEPTS



Concept	Definition	Key Note
Deflation	A persistent fall in the general level of prices.	Opposite of inflation. Bad for the economy leads to fall in consumption, decline in GDP, and removes producers incentive.
Disinflation	Inflation occurring at a slowing pace.	E.g., if last month's inflation was 6% and this month it is 5%, it is called Disinflation. Prices are still rising, just more slowly.

Reflation	Deliberate action of the government to increase the rate of inflation to stimulate the economy.	Usually done to redeem the economy from a deflationary situation.
Stagflation	Both stagnation (very low economic growth) and price rise occur simultaneously.	A rare phenomenon; exception to the Phillips Curve.
Skewflation	Increase in prices of only a few goods or a group of goods, while prices of other goods remain stable.	Different from inflation, which involves a rise in prices of all goods and services. Common in India for food items.
Shrinkflation	The practice of reducing the size of a product while maintaining its sticker price.	A form of hidden inflation. Strategy employed mainly in food and beverage industries to stealthily boost or maintain profit margins in the face of rising input costs.
Greedflation	Corporations use crises (like supply chain disruptions, wars or pandemics) as a pretext to inflate prices excessively and earn abnormal profits.	Results in a Profit-Price Spiral. Distinct from Profit-Push Inflation where prices rise beyond what costs justify.
Peace-Time Inflation	Sustained increase in general price levels during a period of normal peace.	Often triggered by increased government spending on long-term projects, supply-side issues or shifts in monetary policy.
Sporadic Inflation	Prices of only a few commodities in a few regions rise.	Sectional in nature. E.g., rise in food prices due to bad monsoon.

Inflation and Real Interest Rate

Real Interest Rate = Nominal Interest Rate - Inflation

Inflation Premium: An additional amount that investors require on the return of an investment to compensate for the risk of inflation. In order to negate the effect of inflation, investors calculate a 'Real Interest Rate' by adjusting the 'Nominal Interest Rate' for inflation. Accordingly, they charge the borrower on the real interest rate, and not on the nominal interest rate.

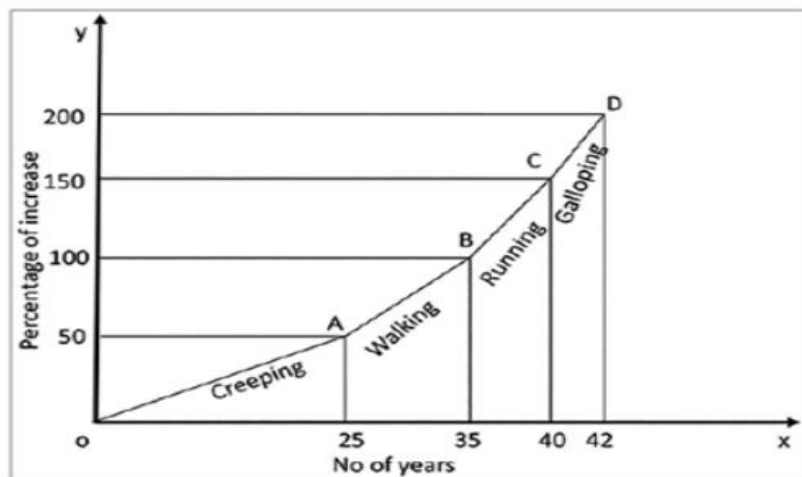
Whom Does Inflation Benefit?

Losers (Harmed by Inflation)	Gainers (Benefit from Inflation)
Lenders / Creditors	Borrowers / Debtors
Bondholders	Bond Issuer

Fixed Income Groups	Businessmen (can raise prices)
Exporters (less competitive due to high domestic prices)	--

Mild inflation (around 2%-4%) is good for the economy as it incentivizes producers to increase production, increases tax income of the government, and creates employment.

TYPES OF INFLATION (BASED ON RATE OF PRICE RISE)



Type	Rate Range	Key Feature
Creeping Inflation (Mild / Low)	2% to 3%	Slow but continuous rise in prices. Manageable and generally considered good for the economy. Producers and traders make reasonable profits, encouraging further investment.
Walking Inflation (Trotting)	3% to 10%	Faster than creeping inflation. The government must treat it as a wake-up call. If unchecked, it may lead to galloping inflation.
Galloping Inflation (Hopping / Running)	10% to 50%	High level of price rise. Businesses and employees' incomes cannot keep pace with costs. Investors refrain from investing.
Hyperinflation	More than 50% per month	Extreme form. The value of domestic currency reduces to almost zero. Usually occurs due to a large increase in money supply. Examples: Zimbabwe (2007-09), Venezuela (2019).

TYPES OF INFLATION (BASED ON CAUSES)

Type	Cause	Key Points
Demand-Pull Inflation	Rise in aggregate demand exceeds aggregate supply.	Caused by: Increased disposable income, expansion in money supply, consumer expectations, fiscal stimulus, growing economy, credit boom, increased marginal propensity to consume.
Cost-Push Inflation	Increased cost of one or more inputs of production.	E.g., rise in wage of labour, higher input prices, exchange rate depreciation, declining productivity, natural disaster, monopoly.
Supply-Shock Inflation	Unexpected or unforeseen sharp fall in supply of commodities.	Causal factors lie out of the control of either the firms or the workers.
Monetary Inflation	Large increase in money supply.	When there is more money in circulation, consumers have more purchasing power, driving up demand and prices. Central banks printing more money contributes to this.
Structural / Bottleneck Inflation	Deficiencies in the economy: inefficient storage/distribution, poor productivity.	These deficiencies lead to supply shortages, raising prices. The government faces a tough time tackling this type.
Profit-Push Inflation	Firms raise prices beyond what costs justify, to protect or increase profit margins.	Contributes to inflation because prices rise not only due to costs but also due to profit motive. E.g., OPEC.
Protein Inflation	Changing dietary habits due to increase in income.	As more people include protein-rich items (milk, pulses, meat, fish), demand and prices of these items rise.

Built-In / Wage-Price Spiral

Caused by a feedback loop of workers demand higher wages leading to businesses raising prices and workers demand higher wages again. Collective bargaining by labour unions can trigger this cycle.



Inflation Based on Government's Reaction

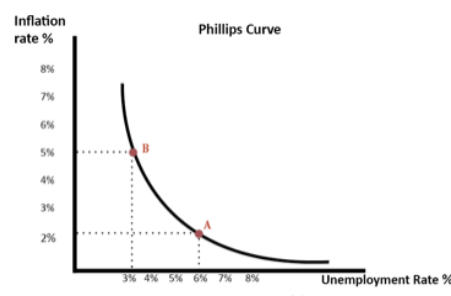
Open Inflation: When the government does not attempt to restrict inflation. In a free market where prices are allowed to take their course, open inflation occurs.

Suppressed Inflation: When the government prevents price rise through price controls, rationing, etc. Also called repressed inflation.

Phillips Curve

It represents the **inverse relationship** between the unemployment rate and inflation. If unemployment decreases, inflation will increase and vice-versa. This curve is named after the economist **A.W. Phillips**, who first put forward this theory in **1958**.

Stagflation is a rare phenomenon and is an exception to the Phillips Curve, it involves both stagnation (very low economic growth) and price rise occurring simultaneously.



Inflation Tax

It refers to an implicit tax caused by the effect of inflation on nominal assets. Printing and injecting more money into the market by the government creates an inflationary situation, which reduces the purchasing power of the money held by individuals. Thus, inflation acts like an explicit tax levied by the government as its effects are similar.

Deflationary Spiral

A deflationary spiral is a situation where falling prices lead to lower production, which leads to lower incomes, which leads to further falling demand and further price falls creating a self-reinforcing cycle of economic decline.

EFFECTS OF INFLATION

Area	Effect
Production and Consumption	A rise in prices leads to a fall in demand. May force a cut down in production. May also lead to a shift in investment from other sectors to those in which price rise has taken place.
Savings	As inflation reduces the value of money, holding money is not prudent. People prefer to deposit money in banks to negate the effect of inflation.
Growth and Employment	In the short run, an increase in inflation is usually accompanied by an increase in economic growth and employment. In the long term, this does not necessarily hold true.
Balance of Payment (BoP)	High prices reduce exports and increase imports. Results in an unfavorable Balance of Payment.

Exchange Rate	High import and low export means high demand for foreign currencies vis-à-vis domestic currency. This depreciates the domestic currency.
Social and Political	High price levels lead to social and political tensions like strikes, dharnas, etc.

MEASURES OF INFLATION

In India, inflation is mainly measured through two price indices: the **Wholesale Price Index (WPI)** and the **Consumer Price Index (CPI)**. Another measure frequently used across the world is the **GDP Deflator**.

Key Concepts

Term	Definition
Price Level	Average price or cost of all goods and services offered for sale in an economy. Measurement of price level allows economists to monitor changes in prices over a period of time.
Price Index	A statistical measure that reflects the average change in prices over time for a basket of goods and services. Expressed with reference to a particular period (the Base Year). A base year is selected and its index is assumed as 100; the price index for other periods is calculated on its basis.
Base Effect	When changes in the CPI in the base month have a considerable impact on changes in YoY inflation, this is referred to as base effect. It refers to the impact that an unusually high or low reference point (the baseline or previous period) has on the calculation of a current percentage change or growth rate.

CONSUMER PRICE INDEX (CPI)

Consumer Price Index measures the average change of prices paid by the final consumer for a basket of goods and services.

$$\text{CPI} = (\text{Weighted Price for Current Year} / \text{Weighted Price for base year}) \times 100$$

CPI (Rural, Urban and Combined) are three new all-India indices. The older indices such as CPI-AL/RL or CPI-IW (Agricultural Labour and Industrial Workers) only covered a segment of population and didn't give a complete countrywide picture. CPI (Rural, Urban and Combined) cover all segments of population. These are published at all-India as well as state-wise levels.

CPI is more suitable for monetary policy and for measuring retail inflation. **Base year revised to 2024**. For compiling CPI, **Modified Laspeyre's Index Formula** is used, which is based on three elements: Base Year Price, Base Year Weight and Current month's price.

New CPI (Base 2024) vs Old CPI (Base 2012)

Feature	New CPI (Base 2024)	Old CPI (Base 2012)
Base Year	2024	2012
Number of Items	358 items (goods: 259→308; services: 40→50)	299 items (288 rural, 281 urban)
New Additions	Rural housing, Online media/OTT services, Value-added dairy products, Barley products, Pen-drive & External Hard disk, Attendant, Babysitter, Exercise equipment, etc.	--
Items Removed	VCR/VCD/DVD player, Radio, Tape recorder, Clothing second-hand, CD/DVD audio/video cassettes, Coir/rope	--
Weight Reference	HCES 2023-24 (Household Consumption Expenditure Survey)	Consumer Expenditure Survey (CES) 2011-12
Classification System	12 Divisions based on COICOP 2018	6 Groups based on COICOP 1999
Data Collection	CAPI (Computer Assisted Personal Interviewing) on tablets	Paper-based schedules
Housing Index	All India, urban and rural (Rural rent included for the first time)	All India and Urban only
Weight of Food & Beverages	36.75%	45.86%
Weightage of Services	50%	40%
Core	58%	47.3%

The revised CPI will include both rural and urban housing inflation, using data from the HCES 2023–24. **E-commerce prices** introduced as an additional market in 12 large towns (population exceeding 25 lakhs). **Online sources** for select services such as airfares and OTT subscriptions. **Administrative data** for standardised services such as rail fares, postal charges and fuel prices.

Key Issues in the Current CPI

- **Exclusion of Free PDS Items:** Free goods distributed under the Public Distribution System are currently excluded as they do not involve direct monetary transactions. This aligns with

international practices and IMF recommendations.

- **Challenges with PDS Inclusion:** Redistribution of weights for free items has occasionally caused inflation spikes. Free items complicate inflation calculation as their inclusion may distort the measure of consumer expenditure.

CPI-IW and CPI-AL

Index	Published By	Purpose	Key Feature
CPI-IW (Industrial Workers)	Labour Bureau	Measures impact of price rise on cost of living for working class families in select industries.	Base year revised from 2001 to 2016. Covers 7 sectors: Factories, Mines, Plantation, Public Motor Transport Undertaking, Electricity Generating & Distribution Establishments, Ports & Docks.
CPI-AL (Agricultural Labourers)	Labour Bureau, Shimla, Ministry of Labour	Measures changes in prices of commodity basket consumed by rural labourers like agri labourers, village/cottage industry labourers.	Used for revising minimum wages for agricultural labourers in different states. CPI-AL is a subject of CPI for Rural Labourers.

WHOLESALE PRICE INDEX (WPI)

The Wholesale Price Index (WPI) is a key macroeconomic indicator used to measure the **average change in the prices of goods traded in bulk at the wholesale level**. It is a **goods-based index**. WPI measures price changes only in physical goods and excludes services such as healthcare, education, and telecom. It captures prices at the **wholesale or producer level**, making it an important indicator of **supply-side inflation**. The revised WPI series uses **2022-23 as the base year**.

Major Components of WPI

Component	Coverage
Manufactured Products (Highest Weight)	Processed foods, textiles, chemicals, machinery, metals, and other manufactured goods.
Primary Articles	Raw and unprocessed items such as food grains, vegetables, oilseeds, and minerals.
Fuel & Power	Energy products like coal, petroleum products, natural gas, and electricity.

Significance of WPI

- **Corporate Contracts:** Used in price escalation clauses to adjust payments in long-term contracts and infrastructure projects.
- **Policy Formulation:** Helps the government design fiscal and trade measures to manage inflation and supply-side pressures.

- **GDP Deflator:** Assists in converting nominal GDP into real GDP by accounting for the impact of inflation.

Key Highlights of WPI (Base Year 2022-23)

- **Expansion of WPI Basket:** Total number of items tracked expanded from **697 to 957 items**.
- **Inclusion of Renewable Energy:** Solar and Wind energy, alongside Nuclear Electricity, integrated under the 'Electricity' Group.
- **Restructuring of Energy Tracking:** Crude Petroleum and Natural Gas shifted from 'Primary Articles' into the 'Fuel and Power' group.
- **Methodological Shift to Gross Value of Output (GVO):** Replaces the older Net Traded Value (GVO + Imports - Exports) methodology used in the 2011-12 series.
- **Introduction of PPI Framework:** The new series will simultaneously release Output Producer Price Index (OPPI), a Trial Input Producer Price Index (IPPI) (manufacturing sector, experimental), and a quarterly Service Producer Price Index covering seven sectors: Banking, Securities Transaction, Insurance, Management of Pension Funds, Railways, Air (Passenger) and Telecom.
- **Divergent Pricing Valuation:** WPI, Output PPI and Service PPI are compiled based on 'Basic Price' (excludes Net Tax and Trade & Transport Margins). Input PPI is compiled using 'Purchaser's Price' since industries source inputs directly from the market.

PRODUCER PRICE INDEX (PPI)

Producer Price Index (PPI) measures the average change over time in the prices received by domestic producers for goods and services they sell. It tracks inflation from the **producer's or seller's perspective**, capturing price changes before they reach consumers. While the CPI measures what consumers pay, PPI measures what manufacturers and service providers receive. PPI serves as an **early indicator of inflationary pressures** and helps assess how input costs are transmitted to final product prices.

Pillar	Definition	Example
Input PPI	Measures the change in prices of raw materials, goods, and services that industries purchase to manufacture their final products.	Cost of steel, rubber and electricity for a car manufacturer.
Output PPI	Measures the change in prices industries receive for the final goods and services they produce.	Factory price of the finished car.
Service PPI	Specifically tracks price changes in the service sector, which makes up over 50% of India's GDP.	Tracks services previously ignored by the old WPI.

WPI vs CPI: COMPARISON

Basis	WPI (2022-23)	CPI Combined (2024)
Base Year	2022-23	2024
Number of Items	Around 957 items	Around 358 items

Services	Not included.	Includes services.
Weightage of Food Products	Food items have lesser weight (around 24.4%).	Food items have greater weight (around 36.75%).
Rate of Inflation	WPI-based inflation is generally low compared to CPI-based inflation.	CPI-based inflation is generally high compared to WPI-based inflation.
Suitability	Less suitable for monetary policy decisions.	More suitable for monetary policy decisions; adopted by RBI as key measure.

WPI vs CPI vs GDP Deflator

Wholesale Price Index (WPI)	Consumer Price Index (CPI)	GDP Deflator
Includes only goods	Includes both goods and services	
Includes capital goods	Does not include capital goods	Includes capital goods
Includes imported items		Excludes imported items
Excludes indirect taxes	Includes indirect taxes	
Office of Economic Adviser (MoCI)	Released by NSO (MoSPI)	
Does not reflect up-to-date expenditure patterns		Reflects up-to-date expenditure patterns
Does not consider services sector	Limited range of goods and services	Enhanced inflation measurement

Why Does Retail Inflation Become Higher Than Wholesale Inflation?

- High transaction costs.
- Weak infrastructure system.
- Weak information system.
- Poor marketing facilities.
- Collusion among traders leading to poor competition.
- Asymmetry in the transmission of price signals from wholesale to retail prices and vice-versa.

Cobweb Phenomenon

Cobweb theory is the idea that **price fluctuations can lead to fluctuations in supply which cause a cycle of rising and falling prices**. The farmers are caught in the cobweb phenomenon when they base their sowing decisions on prices witnessed in the previous marketing period. So, if the farmer observes a higher price for a specific crop in period 't-1', he will opt to produce more of it in period 't'. However, if the production of the crop exceeds market demand, prices fall in period 't', signaling farmers to produce less of the commodity in period 't+1'.

Housing Sector Prices

The **National Housing Bank (NHB)** publishes two housing price indices with FY18 as the base year: **HPI Assessment Price** and **HPI Market Price Quarterly**. HPI Assessment Price is based on the valuation of prices of residential units collected from primary lending institutions. HPI Market Price is based on the market prices for unsold inventories collected from developers. A **composite index** is calculated for 50 cities across India using the population of the cities as weight.

INFLATION-INDEXED BONDS (IIBs)

When the principal amount of a bond is indexed to inflation, it is called an **inflation-indexed bond**. The principal amount rises or falls with the rate of inflation. Regardless of the amount of inflation in the economy, the investor is guaranteed a continuous return.

Formula: Adjusted principal = (Inflation index at a given point of time / Inflation index at the time of deposit) × principal amount. Interest being paid = adjusted principal × coupon rate.

Example: Principal = Rs.100 at 5% coupon rate. If inflation rises 10%, adjusted principal = Rs.110. Interest = Rs.110 × 5% = Rs.5.5 (vs Rs.5 for a fixed deposit). At redemption, adjusted principal or face value, whichever is higher, is paid.

Features of IIBs

- Coupons are paid on a half-yearly basis. The fixed coupon rate is paid on the adjusted principal.
- IIBs are issued for **10 years**.
- As they are G-secs, they are eligible for **SLR status**.
- Minimum investment: Rs. 5,000. Maximum limit: **Rs. 10 lakh per annum for individuals; Rs. 25 lakh per annum for institutions**.
- The inflation-indexed bond was first issued by the **State of Massachusetts** during the American Revolution to combat inflation's effects on consumer goods during the war.
- In India, IIBs were announced in the **2013 budget** mainly to provide an alternative for households to invest in rather than gold, as increasing Current Account Deficit (CAD) driven by higher gold imports was a serious concern.

Why Are IIBs Currently Linked to WPI, Not CPI?

When the bonds were launched in 2013, the **WPI was the prevailing and consistent measure of inflation** and India's monetary policy also targeted WPI for price stability. Although CPI was introduced later, the initial IIBs were tied to the WPI to curb inflation, protect savings, and reduce the current account deficit caused by gold imports driven by negative real interest rates.

The United States, India, Canada, and many other countries now issue Inflation-indexed bonds. There will be rising demand in the coming future as these bonds can hedge inflation risks and protect the capital amount of the investor.

EFFORTS TO CONTROL INFLATION

- **Monetary and Fiscal Policy measures.**
- Advisories to state governments for strict action against **hoarding and black marketing** and to effectively enforce the Essential Commodities Act, 1955 and the Prevention of Black-Marketing and Maintenance of Supplies of Essential Commodities Act, 1980.
- **Higher MSPs** to incentivize production of food items in short supply.
- **Price Stabilization Fund Scheme (PSF)** implemented to control price volatility of agricultural commodities like Potato, Onion and Pulses.
- Enhancement of **buffer stocks** of pulses.
- **Export duties** on certain food items.
- **Essential Commodities Act.**

India's Tomato and Onion Conundrum

Cause	Detail
Seasonality	70% of tomatoes and 70% of onions are grown in Rabi season (prices low from March onwards). By July-August there is an upward trend in prices due to lower supply.
Irregular Shocks	Uncertain/unpredictable weather events. Disease and pests. E.g., in 2023, prolonged excess summer heat led to a high proportion of bad-quality onions, making good-quality onions expensive despite ample stock.
Poor Infrastructure	Lack of proper cold storage and transportation facilities leads to a large percentage of perishable food items getting wasted.

Steps Taken

- **Mission on Integrated Development of Horticulture:** Provides assistance of 50% of total cost of Rs. 1.75 lakh per unit for low-cost onion storage structure (capacity: 25 tonnes each).
- **Scheme for Agriculture Marketing and Infrastructure for rural godowns:** Enables small farmers to enhance their holding capacity to sell produce at remunerative prices.
- **Operation Greens:** Initially focused on integrated development of Tomato, Onion and Potato (TOP) value chain. Provides 50% subsidy on transportation and storage. Later extended to other vegetables and fruits.
- **Kisan Rail and Krishi Udan:** Focused on speedy movement of perishables.
- **Immediate/Temporary steps:** In Aug 2023, GoI imposed a 40% duty on Onion exports.

PREVIOUS YEAR QUESTIONS

1. With reference to the Indian economy, what are the advantages of “Inflation-Indexed Bonds (IIBs)”? (2022)

- (1) The government can reduce the coupon rates on its borrowing by way of IIBs.
- (2) IIBs provide protection to investors from uncertainty regarding inflation.
- (3) The interest received as well as capital gains on IIBs are not taxable.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

2. Which is likely to be the most inflationary in its effect? (2021)

- (a) Repayment of public debt
- (b) Borrowing from the public to finance a budget deficit
- (c) Borrowing from banks to finance a budget deficit
- (d) Creating new money to finance a budget deficit

3. Which is responsible for maintaining price stability by controlling inflation? (2021)

- (a) Department of Consumer Affairs
- (b) Expenditure Management Commission
- (c) Financial Stability and Development Council
- (d) Reserve Bank of India

4. Demand-pull inflation can be caused/increased by: (2020)

- 1. Expansionary policies
- 2. Fiscal stimulus
- 3. Inflation-indexing wages
- 4. Higher purchasing power
- 5. Rising interest rates

Select the correct answer using the code given below.

- (a) 1, 2 and 4 only
- (b) 3, 4 and 5 only
- (c) 1, 2, 3 and 5 only
- (d) 1, 2, 3, 4 and 5

5. Which of the following factors/policies were affecting the price of rice in India in the recent past? (2020)

- 1. Minimum Support Price
- 2. Government's trading
- 3. Government's stockpiling
- 4. Consumer subsidies

Select the correct answer using the code given below:

- (a) 1, 2 and 4 only
- (b) 1, 3 and 4 only
- (c) 2 and 3 only
- (d) 1, 2, 3 and 4

6. Consider the following statements: (2020)

- 1. The weightage of food in the Consumer Price Index (CPI) is higher than that in the Wholesale Price Index (WPI).
- 2. The WPI does not capture changes in the prices of services, which the CPI does.
- 3. The Reserve Bank of India uses WPI as its key measure of inflation to decide changes in policy rates.

Which of the statements given above is/are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

7. With reference to inflation in India, which is correct? (2015)

- (a) Controlling the inflation is the responsibility of the Government of India only
- (b) The Reserve Bank of India has no role in controlling the inflation
- (c) Decreased money circulation helps in controlling the inflation
- (d) Increased money circulation helps in controlling the inflation

8. Economic growth is usually coupled with: (2011)

- (a) Deflation
- (b) Inflation
- (c) Stagflation
- (d) Hyperinflation

9. What is “base effect”? (2011)

- (a) It is the impact of drastic deficiency in supply due to failure of crops.
- (b) It is the impact of the surge in demand due to rapid economic growth.
- (c) It is the impact of the price levels of previous year on the calculation of inflation rate.
- (d) None of the statements (a), (b) and (c) given above is correct in this context.

10. A rise in general level of prices may be caused by:

- 1. An increase in the money supply.
- 2. A decrease in the aggregate level of output.
- 3. An increase in the effective demand.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

Mains Practice Questions

- 1. Explain the relationship between inflation and unemployment with a special reference to the Indian economy. How do policymakers balance these two macroeconomic goals?
- 2. Highlight the effects of sustained high inflation on businesses, consumers, and the overall investment climate in India. Also, suggest measures to mitigate these effects.

GALLANT IAS